

ZIP and GZIP Formats

Aspect	ZIP Format	GZIP Format
Full Form	Zone Information Protocol (commonly just called ZIP archive format)	GNU Zip (developed by GNU Project)
Primary Purpose	File archiving + compression (multiple files and folders in one container)	File/stream compression only (single file at a time)
File Extension	.zip	.gz (or .tar.gz when combined with TAR)
Compression Algorithm	Mostly Deflate (LZ77 + Huffman), but can support others (LZMA, BZIP2)	Always Deflate (LZ77 + Huffman)
Multi-file Support	Yes – can compress multiple files and folders in one archive	No – only one file (for multiple files, TAR is used first)
Structure	Local file headers + compressed data + central directory (index of files)	Header (metadata) + compressed block + trailer (CRC + original size)
Error Checking	Optional per-file CRC-32	Always includes CRC-32 checksum and original file size
Metadata	Stores filename, timestamps, permissions, directory hierarchy	Stores filename, timestamp, compression method, checksum
Encryption Support	Yes (password protection, though weak in standard ZIP)	No built-in encryption (requires external tools)
Compression Ratio	Moderate to good (depends on data, multiple algorithms possible)	Slightly better for single large files (stream-based)
Platform Usage	Cross-platform (Windows, macOS, Linux, etc.), common for file sharing	Predominantly Linux/Unix systems, widely used in web servers & networking
Typical Command	zip archive.zip file1 file2 unzip archive.zip	gzip file.txt → file.txt.gz gunzip file.txt.gz
Applications	Software distribution, backups, document sharing, packaging multiple files	Web page compression (HTTP Content-Encoding: gzip), log file compression, backups